

Coding Challenge: Basic CI/CD Pipeline for a Full-Stack App

 **Scenario:** You are working in a DevOps team and need to automate the deployment of a simple **full-stack application** (built using **.NET Core, React, MongoDB, and SQL Server**) using **GitHub Actions** and **Docker**.

Task Overview

You will:

- ✓ **Containerize** a simple **backend and frontend** application.
- ✓ **Write a basic CI/CD pipeline** using **GitHub Actions** to automate build and deployment.
- ✓ **Deploy locally using Docker Compose** instead of Kubernetes.

 **Time Allocation (120 mins)**

- **30 mins:** Containerizing backend & frontend
 - **40 mins:** Setting up CI/CD pipeline (GitHub Actions)
 - **30 mins:** Deploying & testing locally
 - **20 mins:** Documentation & troubleshooting
-

Task Breakdown

1 Containerize the Application (Docker)

 **Steps:**

- Create a Dockerfile for the **.NET Core API** (backend).
- Create a Dockerfile for the **React frontend**.
- Define a `docker-compose.yml` file to start all services (backend, frontend, and MongoDB).

♦ **Expected Outcome:** Running `docker-compose up` should start the backend, frontend, and database.

2 Set Up CI/CD Pipeline (GitHub Actions)

 **Steps:**

- Create a **GitHub Actions workflow** (`.github/workflows/deploy.yml`) that:
 - **Builds & pushes** Docker images to **Docker Hub**.

- **Runs tests** for .NET Core and React.
- **Deploys the app locally** using `docker-compose up`.

◆ **Expected Outcome:** On pushing to the main branch, the pipeline should **build, test, and deploy** automatically.

3 Deploy and Test Locally

📌 Steps:

- Run `docker-compose up` to ensure the application runs correctly.
- Access the frontend via `http://localhost:3000` and test API calls.

◆ **Expected Outcome:** The **frontend should fetch data from the backend** successfully.

🎯 Submission Instructions

1. Upload:
 - Dockerfile (backend & frontend)
 - `docker-compose.yml`
 - `.github/workflows/deploy.yml`
2. Provide a short **README** with deployment steps.

📊 Self-Evaluation Criteria

Criteria	Excellent 🏆	Satisfactory ✅	Below Expectation ❌
Dockerization	Both backend & frontend properly containerized	Works but inefficiently optimized	Issues in Docker setup
CI/CD Pipeline	Automated, functional pipeline	Works but missing steps	Pipeline fails
Local Deployment	Works smoothly with <code>docker-compose up</code>	Partially functional	Doesn't work